Topic 8 — Nuclear and Atomic Physics

MANDATORY SYLLABUS SCOPE: Covers all dimensions of Nuclear Structure, Binding Energy, Radioactivity (α, β, γ decay), Half-Life and Radioactive Dating, Nuclear Fission (Chain Reactions, Critical Mass, Energy Yields, Fissile vs Fertile Fuels), Nuclear Reactor Components (Fuel, Moderator, Control Rods, Coolant, Shielding), Nuclear Fusion (Solar/Stellar Energy, Thermonuclear Weapons, Hydrogen Bomb, Tokamak Confinement), India's Three-Stage Nuclear Power Programme (PHWR, PFBR/FBR, AHWR), Nuclear Fuel Reserves in India (Monazite, Thorium vs Uranium Deposits, LEU vs HEU), Radiation Applications and Detection Instruments (Scintillation Counter, Cyclotron), and International Treaties (NPT, CTBT). Fully synchronized with Ghatna Chakra Physics, NCERT Class 12, and latest 2021–2025 State PCS PSC papers.

Current Affairs (2024–2026 Focus)

1. **India's Three-Stage Nuclear Programme & Kalpakkam PFBR (High-Yield Science Fact):** - India's nuclear doctrine, drafted by **Dr. Homi Jehangir Bhabha**, is designed to exploit India's vast **Thorium** reserves:
    - **Stage 1:** Pressurized Heavy Water Reactors (PHWR) fuelled by **Natural Uranium ($\text{U-238} + 0.7\%\ \text{U-235}$)**, producing electricity and byproduct **Plutonium-239 (Pu-239)**.
    - **Stage 2:** Fast Breeder Reactors (FBR) fuelled by **Plutonium-239** with a Thorium-232 blanket, breeding fissile **Uranium-233 (U-233)**. Core loading commenced at the $500\ \text{MWe}$ **Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, Tamil Nadu**.
    - **Stage 3:** Advanced Heavy Water Reactors (AHWR) fuelled by **Thorium-232 / Uranium-233**, unlocking energy independence.
2. **Monazite Placer Sands & Legal Monopoly in India (IAS Pre 2022 Fact):** - **Monazite** is a complex reddish-brown phosphate mineral of Thorium and rare-earth elements $[(\text{Ce, La, Nd, Th, Y})\text{PO}_4]$. - Coastal beach placers occur in **Kerala, Tamil Nadu, Odisha, Andhra Pradesh, Maharashtra, and Gujarat** (estimated at $13.15\ \text{million tonnes}$ by AMD). - **Legal Fact:** Monazite is a prescribed strategic substance under the Atomic Energy Act, 1962. **Private entities are strictly prohibited from processing or exporting monazite**. Only government undertakings—**IREL (Indian Rare Earths Limited)** under DAE and **KMML (Kerala Minerals and Metals Ltd)**—are legally authorized to mine, process, and extract thorium compounds.
3. **Uranium Enrichment Levels for Civil vs Military Use (IAS Pre 2023 Fact):** - **Natural Uranium:** Contains only 0.7% **U-235**; the remaining 99.3% is non-fissile U-238. - **Low Enriched Uranium (LEU):** Enriched up to $3\% - 5\%$ **U-235**, which is standard for commercial civil nuclear power reactors. - **Highly Enriched Uranium (HEU):** Enriched to $> 20\%$ U-235. **Weapons-grade uranium requires $\geq 90\%$ U-235** to achieve compact critical mass. - **Common Trap Fact:** Commercial electricity DOES NOT require 60% enriched uranium (IAS Pre 2023 statement debunked). India derives 43.5% of its electrical power from coal, not uranium.

4. **Narora Atomic Power Station (NAPS, Uttar Pradesh Fact):** - Located in **Narora, Bulandshahr district, Uttar Pradesh** on the banks of the Ganga River. Operates two indigenous PHWR units of $220\ \mathrm{MWe}$ **each** using natural uranium and heavy water.

5. **International Non-Proliferation Treaties (NPT & CTBT Must-Score):** - **NPT (Non-Proliferation Treaty):** Opened for signature in 1968, in force 1970, extended indefinitely in 1995 (191 **parties**). Recognizes 5 Nuclear-Weapon States (tested before 1 Jan 1967: US, Russia, UK, France, China). **India, Pakistan, Israel, and South Sudan have never signed**. North Korea announced withdrawal in 2003. - **CTBT (Comprehensive Nuclear-Test-Ban Treaty):** Adopted 1996; requires ratification by all 44 Annex-2 states. Not in force because India, Pakistan, and North Korea have not signed, and US, China, Egypt, Iran, and Israel have not ratified. Russia revoked ratification in 2023.

Detailed In-Depth Core Theory

8.1 Nuclear Energetics, Mass Defect and Fission Physics

1. **Binding Energy and Mass Defect:** - The mass of an atomic nucleus is always strictly less than the sum of the individual rest masses of its constituent protons and neutrons. - Mass defect $\Delta m = [Z m_p + (A-Z)m_n] - M_{\text{nucleus}}$. - Nuclear Binding Energy: $\Delta E = \Delta m \cdot c^2$. A mass defect of $1\ \text{amu} = 931.5\ \text{MeV}$.
2. **Mechanism of Fission:** - When a slow thermal neutron strikes a $^{235}_{92}\text{U}$ nucleus, a compound nucleus $^{236}_{92}\text{U}^*$ forms in an excited state, oscillates, and undergoes asymmetric fission:

$$^{235}_{92}\text{U} + ^{1}_{0}n \rightarrow ^{236}_{92}\text{U}^* \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3^{1}_{0}n + 200\ \text{MeV}$$

   - Fast neutrons emitted have kinetic energy $\sim 2\ \text{MeV}$. Thermal reactors require a moderator to slow them down to $0.025\ \text{eV}$ (where the fission cross-section of U-235 is 1000 times higher).

8.2 Reactor Components and Safety Engineering

1. **Moderation Mechanics:** - Slowing down occurs through elastic collisions with light nuclei. The closer the mass of the moderator nucleus to that of a neutron ($m_n = 1$), the greater the energy transfer per collision. - Heavy water (D_2O) is optimal because Deuterium (^2H) has mass $\sim 2\ \text{amu}$ and almost zero neutron capture.
2. **Reactivity Control:** - Multiplication factor $k = \frac{\text{Neutrons produced in generation } n+1}{\text{Neutrons absorbed in generation } n}$. - $k = 1$: Critical (steady power output). - $k < 1$: Subcritical (reaction dies out). - $k > 1$: Supercritical (power escalates exponentially). - Cadmium (^{113}Cd) and Boron (^{10}B) have astronomical neutron capture cross-sections, quenching reactivity when inserted.

8.3 Thermonuclear Fusion & Stellar Nucleosynthesis

1. **Thermonuclear Conditions:** - Overcoming electrostatic Coulomb repulsion between two positively charged nuclei requires kinetic energies corresponding to temperatures of $10^7 - 10^8\ \text{K}$.
2. **Proton-Proton Chain in the Sun:**

$$4\,^{1}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + 2e^+ + 2\nu_e + 26.7\ \text{MeV}$$

3. **Tokamak Magnetic Confinement:** - Plasma at $10^8\ \mathrm{K}$ cannot touch any material container. Tokamaks use powerful superconducting magnetic coils to suspend plasma in a toroidal vacuum vessel.

8.4 India's Strategic Nuclear Minerals & Legal Framework

1. **Thorium Exploitation:** - India possesses 25% of the world's thorium reserves. - In Stage 2 FBRs, fast neutrons transmute non-fissile Thorium into fissile U-233:

$$^{232}_{90}\mathrm{Th} + ^{1}_{0}n \rightarrow ^{233}_{90}\mathrm{Th} \xrightarrow{\beta^-} ^{233}_{91}\mathrm{Pa} \xrightarrow{\beta^-} ^{233}_{92}\mathrm{U}$$

2. **Statutory Control:** - Under the Atomic Energy Act, 1962, all exploration, extraction, processing, and export of uranium and thorium are exclusively reserved for Central Government statutory bodies.

Bilingual Terminology -- Nuclear and Atomic Physics

English Term	Hindi Term	Key Anchor
Atom (परमाणु)	परमाणु	Smallest unit of element that retains chemical properties
Nucleus (नाभिक)	नाभिक	Core of atom; protons + neutrons; positively charged; tiny (10^-15 m)
Proton (प्रोटॉन)	प्रोटॉन	Positive charge (+1); in nucleus; mass = 1.67 x 10^-27 kg
Neutron (न्यूट्रॉन)	न्यूट्रॉन	No charge; in nucleus; slightly heavier than proton
Electron (इलेक्ट्रॉन)	इलेक्ट्रॉन	Negative charge (-1); orbits nucleus; mass ~1/1836 of proton
Atomic Number (Z) (परमाणु क्रमांक)	परमाणु संख्या	Number of protons; defines the element
Mass Number (A) (द्रव्यमान संख्या)	द्रव्यमान संख्या	Protons + Neutrons (nucleons)
Isotope (समस्थानिक)	समस्थानिक	Same Z, different A; same element, different neutrons; e.g., U-235 and U-238
Radioactivity (रेडियोधर्मिता)	रेडियोधर्मिता	Spontaneous emission of radiation from unstable nuclei; discovered by Becquerel (1896)
Alpha Particle (अल्फा कण)	अल्फा कण	Helium nucleus (2p + 2n); least penetrating; stopped by paper
Beta Particle (बीटा कण)	बीटा कण	High-speed electron/positron; more penetrating; stopped by aluminium sheet
Gamma Ray (गामा किरण)	गामा किरण	Electromagnetic radiation; most penetrating; stopped by thick lead/concrete
Half-Life (अर्ध आयु)	अर्ध-जीवन	Time for half the radioactive atoms to decay
Nuclear Fission (नाभिकीय विखण्डन)	परमाणु विखंडन	Heavy nucleus splits; e.g., U-235 + neutron; huge energy; basis of atom bomb

English Term	Hindi Term	Key Anchor
Nuclear Fusion (नाभिकीय संलयन)	परमाणु संलयन	Light nuclei combine; e.g., H+H = He; basis of H-bomb; powers the Sun
Chain Reaction (श्रृंखला अभिक्रिया)	श्रृंखला अभिक्रिया	Fission releases neutrons -> more fission -> self-sustaining; critical mass needed

One-Liner Revision -- Nuclear and Atomic Physics

#	Fast Fact
1	Radioactivity discovered by = **Henri Becquerel (1896)**; Nobel 1903 (with Curies)
2	Marie Curie = discovered Polonium and Radium; 2 Nobel Prizes (1903 Physics, 1911 Chemistry)
3	Rutherford's nuclear model = nucleus-centred atom; alpha scattering experiment (1909)
4	Bohr's model = electrons in fixed orbits (energy shells); explains hydrogen spectrum
5	Electron discovered = **J.J. Thomson (1897)**; cathode ray tube experiment
6	Proton discovered = **Ernest Rutherford (1917)**; Nobel 1908
7	Neutron discovered = **James Chadwick (1932)**; Nobel 1935
8	Positron (anti-electron) discovered = **Carl Anderson (1932)**; Nobel 1936
9	Atomic number (Z) = number of **protons**; unique to each element
10	Mass number (A) = protons + neutrons; isotopes have same Z, different A
11	Alpha particles: stopped by **paper / skin**; most ionising; dangerous if inhaled
12	Beta particles: stopped by **aluminium sheet** (few mm)
13	Gamma rays: stopped by **thick lead** (several cm) or thick concrete
14	Half-life of C-14 = **5730 years** (radiocarbon dating of organic matter)
15	Half-life of U-235 = 700 million years; U-238 = 4.5 billion years
16	Nuclear fission: U-235 + neutron -> Ba + Kr + 3 neutrons + 200 MeV
17	Critical mass = minimum fissile material for self-sustaining chain reaction
18	Atom bomb (Little Boy) = U-235; (Fat Man) = Pu-239 (Nagasaki/Hiroshima, 1945)
19	Nuclear fusion: D + T -> He-4 + neutron + 17.6 MeV; requires >10^7 K
20	Hydrogen bomb (thermonuclear bomb) = fusion bomb; triggered by fission bomb
21	Einstein's mass-energy equivalence: E = mc^2
22	Nuclear reactor: controlled fission; moderator (heavy water/graphite) slows neutrons
23	India's 1st nuclear test = **Pokhran (1974)**; "Smiling Buddha"

#	Fast Fact
24	India's 2nd nuclear test = **Pokhran-II (1998)**; "Operation Shakti"
25	India's nuclear doctrine = **No First Use (NFU)** + credible minimum deterrence
26	Chernobyl disaster (1986) = RBMK reactor explosion; Ukraine (Soviet Union)
27	Fukushima disaster (2011) = tsunami-induced meltdown; Japan
28	ITER project = International Thermonuclear Experimental Reactor; France; fusion experiment
29	Radiocarbon dating = measures C-14 decay; used for dating up to ~50,000 years
30	X-rays discovered = **Wilhelm Roentgen (1895)**; Nobel 1901 (first ever)